DOE SBIR/STTR FY26 PHASE I · GENESIS MISSION · TOPIC 1

Bibliography and References Cited

Optional upload. It draws the line between prior art and the PlastiBioFuel contribution, and it shows the UNT partner's published record in enzyme immobilization.

6	3	1
PEER-REVIEWED SOURCES	WITH DR. MA AS AUTHOR	PROVISIONAL PATENT

ENZYMATIC DEPOLYMERIZATION OF PET

1 Lu, H.; Diaz, D. J.; Czarnecki, N. J.; Zhu, C.; Kim, W.; Shroff, R.; Acosta, D. J.; Alexander, B. R.; Cole, H. O.; Zhang, Y.; Lynd, N. A.; Ellington, A. D.; Alper, H. S. **Machine learning-aided engineering of hydrolases for PET depolymerization.** *Nature* 2022, 604 (7907), 662-667.
DOI 10.1038/s41586-022-04599-z · PMID 35478237

> Source of FAST-PETase and of the machine-learning method used to engineer it. PlastiBioFuel did not develop this enzyme and makes no claim to it. Cited to fix the boundary between prior art and the PlastiBioFuel contribution, and because it establishes that machine learning is already productive in this system at the molecular scale. The proposed Phase I work moves that method up one level, from residue selection to reactor operating conditions.

2 Tournier, V.; Topham, C. M.; Gilles, A.; David, B.; Folgoas, C.; Moya-Leclair, E.; Kamionka, E.; Desrousseaux, M.-L.; Texier, H.; Gavalda, S.; Cot, M.; Guemard, E.; Dalibey, M.; Nomme, J.; Cioci, G.; Barbe, S.; Chateau, M.; Andre, I.; Duquesne, S.; Marty, A. **An engineered PET depolymerase to break down and recycle plastic bottles.** *Nature* 2020, 580 (7802), 216-219.
DOI 10.1038/s41586-020-2149-4 · PMID 32269349

> Benchmark for enzymatic PET depolymerization productivity and for the monomer-to-resin route. Establishes the competing commercial pathway that returns terephthalate to virgin-equivalent PET rather than to fuel.

3 Yoshida, S.; Hiraga, K.; Takehana, T.; Taniguchi, I.; Yamaji, H.; Maeda, Y.; Toyohara, K.; Miyamoto, K.; Kimura, Y.; Oda, K. **A bacterium that degrades and assimilates poly(ethylene terephthalate).** *Science* 2016, 351 (6278), 1196-1199.
DOI 10.1126/science.aad6359 · PMID 26965627

> Origin of the PETase and MHETase system, and the source of the terephthalic acid and ethylene glycol product pair that PlastiBioFuel intends to route to ethanol.

ENZYME IMMOBILIZATION IN COVALENT ORGANIC FRAMEWORKS

4 Wang, X.; Lan, P. C.; **Ma, S.** **Metal-Organic Frameworks for Enzyme Immobilization: Beyond Host Matrix Materials.** *ACS Central Science* 2020, 6 (9), 1497-1506.
DOI 10.1021/acscentsci.0c00687 · PMID 32999925

> Authored from the Department of Chemistry, University of North Texas. Dr. Shengqian Ma is the PlastiBioFuel subaward lead for framework and immobilization chemistry under Sponsored Research Agreement SRA1016.

5 Li, M.; Qiao, S.; Zheng, Y.; Andaloussi, Y. H.; Li, X.; Zhang, Z.; Li, A.; Cheng, P.; **Ma, S.**; Chen, Y. **Fabricating Covalent Organic Framework Capsules with Commodious Microenvironment for Enzymes.** *Journal of the American Chemical Society* 2020, 142 (14), 6675-6681.
DOI 10.1021/jacs.0c00285 · PMID 32197569

> Covalent organic framework capsules as enzyme hosts that preserve conformational freedom and mass transfer. This is the physical basis for the near-complete loading with retained catalytic activity that PlastiBioFuel measures in its own framework.

6 Zhang, S.; Zheng, Y.; An, H.; Aguila, B.; Yang, C.-X.; Dong, Y.; Xie, W.; Cheng, P.; Zhang, Z.; Chen, Y.; **Ma, S.** **Covalent Organic Frameworks with Chirality Enriched by Biomolecules for Efficient Chiral Separation.** *Angewandte Chemie International Edition* 2018, 57 (51), 16754-16759.
DOI 10.1002/anie.201810571 · PMID 30359485

> Covalent immobilization of enzymes and other biomolecules into covalent organic frameworks with function retained, from the same group.

PLASTIBIOFUEL INTELLECTUAL PROPERTY

7 Ward, M. D. **U.S. Provisional Patent Application 63/848,456**, filed July 2025.

> Covers covalent organic framework immobilization, spiral-flow reactor geometry, and the integrated depolymerization-to-ethanol process.

VERIFICATION

> References 1 through 6 were retrieved from PubMed and verified on August 22, 2026.

PlastiBioFuel LLC · michael@plastibiofuel.com · 817-319-7383 UEI KQAWZ54RDUM8 · CAGE 175D6